

Knowledge Base Rules (from knowledge_base_rules.csv)

The table below is a Markdown representation of the original CSV file.

Rule	Type	IF (natural language)	THEN (natural language)	Propositional logic	Predicate logic (patient x)
R1	Standard	ML risk probability ≥ 0.35	elevated_cardiac_risk	HighRiskProb $\rightarrow$ ElevatedCardiacRisk	$\forall x (\text{RiskProb}(x) \geq 0.35 \rightarrow \text{ElevatedCardiacRisk}(x))$
R2	Standard	elevated_cardiac_risk AND chest_pain = asymptomatic	silent_ischemia_suspected AND urgent_cardiology_review	ElevatedCardiacRisk $\wedge$ Asymptomatic $\rightarrow$ SilentIschemia $\wedge$ UrgentCardiologyReview	" $\forall x (\text{ElevatedCardiacRisk}(x) \wedge \text{ChestPain}(x, \text{asymptomatic}) \rightarrow \text{SilentIschemia}(x) \wedge \text{UrgentCardiologyReview}(x))$ "
R3	Standard	exercise_angina = true AND oldpeak > 1.0	inducible_ischemia AND recommend_cardiology	ExerciseAngina $\wedge$ OldpeakGt1 $\rightarrow$ InducibleIschemia $\wedge$ RecommendCardiology	$\forall x (\text{ExerciseAngina}(x) \wedge \text{Oldpeak}(x) > 1.0 \rightarrow \text{InducibleIschemia}(x) \wedge \text{RecommendCardiology}(x))$
R4	Standard	number_of_vessels (ca) ≥ 1	significant_vessel_disease AND recommend_admission	VesselsGe1 $\rightarrow$ SignificantVesselDisease $\wedge$ RecommendAdmission	$\forall x (\text{CA}(x) \geq 1 \rightarrow \text{SignificantVesselDisease}(x) \wedge \text{RecommendAdmission}(x))$
R5	Standard	thalassemia = reversible defect	reversible_perfusion_defect AND recommend_cardiology	ReversibleDefect $\rightarrow$ ReversiblePerfusionDefect $\wedge$ RecommendCardiology	" $\forall x (\text{Thal}(x, \text{reversible_defect}) \rightarrow \text{ReversiblePerfusionDefect}(x) \wedge \text{RecommendCardiology}(x))$ "
R6	Standard	resting_blood_pressure ≥ 180	hypertensive_urgency AND control_bp_first	BpGe180 $\rightarrow$ HypertensiveUrgency $\wedge$ ControlBpFirst	$\forall x (\text{RestingBP}(x) \geq 180 \rightarrow \text{HypertensiveUrgency}(x) \wedge \text{ControlBpFirst}(x))$
R7	Standard	age ≥ 65 AND elevated_cardiac_risk	high_priority_elderly	AgeGe65 $\wedge$ ElevatedCardiacRisk $\rightarrow$ HighPriorityElderly	$\forall x (\text{Age}(x) \geq 65 \wedge \text{ElevatedCardiacRisk}(x) \rightarrow \text{HighPriorityElderly}(x))$
R8	Safety override	kidney_function = impaired	renal_impairment AND avoid_contrast_angiography	KidneyImpaired $\rightarrow$ RenalImpairment $\wedge$ AvoidContrastAngiography	" $\forall x (\text{KidneyFunction}(x, \text{impaired}) \rightarrow \text{RenalImpairment}(x) \wedge \text{AvoidContrastAngiography}(x))$ "
R9	Safety override	on_anticoagulant = true AND (recommend_admission OR recommend_cardiology)	bleeding_risk AND avoid_thrombolysis	OnAnticoagulant $\wedge$ (RecommendAdmission $\vee$ RecommendCardiology) $\rightarrow$ BleedingRisk $\wedge$ AvoidThrombolysis	$\forall x (\text{OnAnticoagulant}(x) \wedge (\text{RecommendAdmission}(x) \vee \text{RecommendCardiology}(x)) \rightarrow \text{BleedingRisk}(x) \wedge \text{AvoidThrombolysis}(x))$
R10	Standard	NOT elevated_cardiac_risk AND ca = 0 AND oldpeak ≤ 1.0 AND exercise_angina = false	low_risk_profile AND safe_for_discharge	$\neg \text{ElevatedCardiacRisk} \wedge \text{CaEq0} \wedge \text{OldpeakLe1} \wedge \neg \text{ExerciseAngina} \rightarrow \text{LowRiskProfile} \wedge \text{SafeForDischarge}$	$\forall x (\neg \text{ElevatedCardiacRisk}(x) \wedge \text{CA}(x) = 0 \wedge \text{Oldpeak}(x) \leq 1.0 \wedge \neg \text{ExerciseAngina}(x) \rightarrow \text{LowRiskProfile}(x) \wedge \text{SafeForDischarge}(x))$
R11	Standard	cholesterol ≥ 240	hyperlipidemia AND recommend_statin	CholGe240 $\rightarrow$ Hyperlipidemia $\wedge$ RecommendStatin	$\forall x (\text{Chol}(x) \geq 240 \rightarrow \text{Hyperlipidemia}(x) \wedge \text{RecommendStatin}(x))$
R12	Standard	fasting_blood_sugar > 120 (fbs = true)	dysglycemia AND recommend_diabetes_workup	FbsTrue $\rightarrow$ Dysglycemia $\wedge$ RecommendDiabetesWorkup	$\forall x (\text{FBS}(x) = \text{true} \rightarrow \text{Dysglycemia}(x) \wedge \text{RecommendDiabetesWorkup}(x))$